

Physics 523 HW 4

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I use the notation in my Real Scalar Field & QED Notes. I usually use p for the magnitude of the momentum 3-vector, $p = |\mathbf{p}|$, so $p^2 = \mathbf{p} \cdot \mathbf{p}$. But for this homework I will use p^2 for the 4-vector dot product: $p^2 = \vec{p} \cdot \vec{p} = \vec{p}^2 = (p^0)^2 - \mathbf{p}^2$. Also I use ϕ for interacting scalar fields.

Part I

1

1. SUPERFICIAL DEGREE OF DIVERGENCE

The Lagrange density in d dimensions has $[\mathcal{L}] = d$ energy units. The scalar mass term is proportional to $m^2\phi^2$, so

$$d = [m^2\phi^2] = [m^2] + [\phi^2] = 2[m] + 2[\phi] = 2 + 2[\phi]. \quad (1)$$

$$\boxed{[\phi] = \frac{d}{2} - 1} \quad (2)$$

The fermion (I only consider spin-1/2) mass term is proportional to $m\bar{\psi}\psi$.

$$d = [m\bar{\psi}\psi] = [m] + [\bar{\psi}\psi] = 1 + 2[\psi], \quad [\bar{\psi}] = [\psi] \quad (3)$$

$$\boxed{[\psi] = \frac{d-1}{2}} \quad (4)$$

The free boson (boson always means spin 1 here) term is proportional to $F^2 = F^{\alpha\beta}F_{\alpha\beta}$.

$$F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha \quad (5)$$

$$[x] = -1 \implies [\partial] = \left[\frac{\partial}{\partial x} \right] = 1 \quad (6)$$

$$d = [F^2] = 2[F] = 2[\partial A] = 2[\partial] + 2[A] = 2 + 2[A] \quad (7)$$

$$2[A] = d - 2 \quad (8)$$

$$\boxed{[A] = \frac{d}{2} - 1 = [\phi]} \quad (9)$$

The relevant propagators are

$$\vec{A} \sim -\frac{i\eta_{\mu\nu}}{p^2 + i\epsilon}, \quad \phi \sim \frac{i}{p^2 - m^2 + i\epsilon}, \quad \psi \sim i\frac{\not{p} + m}{p^2 - m^2 + i\epsilon}. \quad (10)$$

I only track momentum factors for power counting.

$$\boxed{\vec{A} \sim \frac{1}{p^2} \sim \phi}, \quad \boxed{\psi \sim \frac{1}{p}}. \quad (11)$$

D is the superficial degree of divergence. $\mathcal{N}/\mathcal{D}$ is the number of momenta factors in the numerator/denominator.

$$D = \mathcal{N} - \mathcal{D} \quad (12)$$

$$\mathcal{D} = I_f + 2I_b + 2I_s \quad (13)$$

$I_f/I_b/I_s$ are the number of internal fermion/boson/scalar lines.

$$\mathcal{N} = dL + \sum_{n=1}^N \delta_n V_n \quad (14)$$

L is the number of loops (each contributes $d^d p$); there are N vertex types in $\mathcal{L}$, V_n is the total number of type n vertices in a diagram, and δ_n is the number of derivatives in vertex n .

$$D = dL + \sum_{n=1}^N \delta_n V_n - I_f - 2I_b - 2I_s \quad (15)$$

The total number of internal lines is

$$I = I_f + I_b + I_s. \quad (16)$$

$$D = dL + \sum_{n=1}^N \delta_n V_n - I - I_b - I_s \quad (17)$$

The total number of vertices in a diagram is

$$V = \sum_{n=1}^N V_n. \quad (18)$$

$$L = I - V + 1, \quad (19)$$

Each internal line contributes $d^d p$, each vertex Dirac delta kills a $d^d p$, and one overall momentum conserving delta remains (+1).

$$D = d - dV + \sum_{n=1}^N \delta_n V_n + (d-1)I - I_b - I_s \quad (20)$$

F_n is the number of fermion fields in vertex n ,

$$\sum_{n=1}^N F_n V_n = 2I_f + f, \quad (21)$$

where f is the number of external fermions. The factor of 2 is there because each internal line connects 2 vertices.

$$I_f = \frac{1}{2} \left(\sum_{n=1}^N F_n V_n - f \right) \quad (22)$$

$$I_b = \frac{1}{2} \left(\sum_{n=1}^N B_n V_n - b \right), \quad I_s = \frac{1}{2} \left(\sum_{n=1}^N S_n V_n - s \right) \quad (23)$$

B_n/S_n are the number of bosons/scalars in a type n vertex, and b/s are the number of external boson/scalar fields in the diagram.

$$D = d - dV + \sum_{n=1}^N \delta_n V_n + (d-1)I_f + (d-2)(I_b + I_s) \quad (24)$$

$$D = d - \frac{d-1}{2}f - \left(\frac{d}{2} - 1\right)(b+s) - dV + \sum_{n=1}^N \delta_n V_n + \frac{d-1}{2} \sum_{n=1}^N F_n V_n + \left(\frac{d}{2} - 1\right) \sum_{n=1}^N (B_n + S_n) V_n \quad (25)$$

$$V = \sum_{n=1}^N V_n \quad (26)$$

$$D = d - \frac{d-1}{2}f - \left(\frac{d}{2} - 1\right)(b+s) + \sum_{n=1}^N \left[\delta_n - d + \frac{d-1}{2}F_n + \left(\frac{d}{2} - 1\right)(B_n + S_n) \right] V_n \quad (27)$$

$$[\psi] = \frac{d-1}{2}, \quad [\phi] = [A] = \frac{d}{2} - 1 \quad (28)$$

$$D = d - \frac{d-1}{2}f - \left(\frac{d}{2} - 1\right)(b+s) + \sum_{n=1}^N (\delta_n - d + [\psi]F_n + [A]B_n + [\phi]S_n) V_n \quad (29)$$

Schematically a vertex in $\mathcal{L}$ is $g\phi^S A^B \psi^F \partial^\delta$, where g is the coupling constant.

$$[\mathcal{L}] = [g\phi^S A^B \psi^F \partial^\delta], \quad d = [\mathcal{L}], \quad [\partial] = 1 \quad (30)$$

$$d = [g\phi^S A^B \psi^F \partial^\delta] = [g] + [\phi]S + [A]B + [\psi]F + [\partial]\delta \quad (31)$$

$$- [g] = \delta - d + [\psi]F + [A]B + [\phi]S \quad (32)$$

$$D = d - \frac{d-1}{2}f - \left(\frac{d}{2} - 1\right)(b+s) - \sum_{n=1}^N [g_n] V_n \quad (33)$$

2. RENORMALIZABILITY

By definition renormalizable theories have a finite number of counterterms. Green's functions are classified by their external fields. In any diagram we can take $V_n \rightarrow \infty$. If any $[g_n] < 0$ in Eq. 33 there will be an infinite number of Green's function types with different positive values of D and each will require a counterterm. The number of counterterms is then infinite and the theory is not renormalizable. In a renormalizable theory all couplings must satisfy $[g_n] \geq 0$. Symmetries could remove some divergences but the argument still holds.

s	D
1	4
2	2
3	0
$s > 3$	$D < 0$

TABLE I. Superficial degree of divergence for different s -point Green's functions. All loop diagrams with $s > 3$ are superficially UV finite.

3. ϕ^3 THEORY IN 6D

The interaction Lagrangian has 1 term

$$\mathcal{L}_I = g\phi^3. \quad (34)$$

Eq. 33 becomes

$$D = 6 - 2s - [g]V. \quad (35)$$

Eq. 32 yields

$$[g] = 6 - [\phi]S = 0, \quad [\phi] = 2, \quad S = 3. \quad (36)$$

$$D = 6 - 2s \quad (37)$$

In Table I I show the superficial degree of divergence for different Green's functions.

Part II

2

4. LAGRANGIAN

The bare (B) Lagrangian is

$$\mathcal{L} = -\frac{\phi_B \square \phi_B}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B}{3!} \phi_B^3. \quad (38)$$

I renormalize.

$$\phi_B = \sqrt{Z_\phi} \phi, \quad m_B = \sqrt{Z_m} m, \quad g_B = Z_g g. \quad (39)$$

$$\mathcal{L} = -Z_\phi \frac{\phi \square \phi}{2} - Z_m Z_\phi \frac{m^2 \phi^2}{2} - Z_g Z_\phi^{3/2} \frac{g}{3!} \phi^3 \quad (40)$$

$$Z_1 = Z_g Z_\phi^{3/2} \quad (41)$$

$$\mathcal{L} = -Z_\phi \frac{\phi \square \phi}{2} - Z_m Z_\phi \frac{m^2 \phi^2}{2} - Z_1 \frac{g}{3!} \phi^3 \quad (42)$$

$$Z_\phi = 1 + \delta_\phi, \quad Z_m = 1 + \delta_m, \quad Z_1 = 1 + \delta_1 \quad (43)$$

$$\mathcal{L} = -(1 + \delta_\phi) \frac{\phi \square \phi}{2} - (1 + \delta_m)(1 + \delta_\phi) \frac{m^2 \phi^2}{2} - (1 + \delta_1) \frac{g}{3!} \phi^3 \quad (44)$$

$$\mathcal{L} = -(1 + \delta_\phi) \frac{\phi \square \phi}{2} - (1 + \delta_\phi + \delta_m + \delta_m \delta_\phi) \frac{m^2 \phi^2}{2} - (1 + \delta_1) \frac{g}{3!} \phi^3 \quad (45)$$

$$\mathcal{L} = -\frac{\phi \square \phi}{2} - \frac{m^2 \phi^2}{2} - \frac{g}{3!} \phi^3 - \delta_\phi \frac{\phi \square \phi}{2} - (\delta_\phi + \delta_m + \delta_m \delta_\phi) \frac{m^2 \phi^2}{2} - \delta_1 \frac{g}{3!} \phi^3 \quad (46)$$

The free Lagrangian is

$$\mathcal{L}_0 = -\frac{\phi \square \phi}{2} - \frac{m^2 \phi^2}{2}. \quad (47)$$

$$2\mathcal{L}_0 = -\phi(\square + m^2)\phi \quad (48)$$

The interaction Lagrangian is

$$3!\mathcal{L}_1 = -g\phi^3. \quad (49)$$

The counterterm Lagrangian is

$$\mathcal{L}_C = -\delta_\phi \frac{\phi \square \phi}{2} - (\delta_\phi + \delta_m + \delta_m \delta_\phi) \frac{m^2 \phi^2}{2} - \delta_1 \frac{g}{3!} \phi^3. \quad (50)$$

$$\mathcal{L}_C = -\frac{1}{2} \phi [\delta_\phi \square + (\delta_\phi + \delta_m + \delta_m \delta_\phi) m^2] \phi - \delta_1 \frac{g}{3!} \phi^3 \quad (51)$$

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_C \quad (52)$$

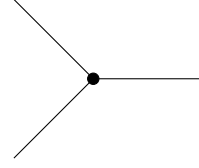


FIG. 1. Interaction vertex with factor $-ig$.

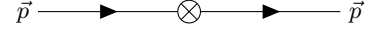


FIG. 2. Counterterm interaction vertex with factor $i[p^2 \delta_\phi - (\delta_\phi + \delta_m) m^2]$.

5. ONE LOOP LAGRANGIAN

I go to one loop order.

$$\mathcal{L}_C = -\frac{1}{2} \phi [\delta_\phi \square + (\delta_\phi + \delta_m) m^2] \phi - \delta_1 \frac{g}{3!} \phi^3 \quad (53)$$

Plugging this into Eq. 52 gives the one loop Lagrangian.

6. FEYNMAN RULES

The propagator is

$$\phi \sim \frac{i}{p^2 - m^2 + i\epsilon}. \quad (54)$$

I show the interaction vertex in Fig. 1, and the counterterm interaction vertices in Figs. 2 & 3.

7. DIMREG

The Euler-Mascheroni constant is $\gamma \approx 0.577216$. I introduce the scale $g \rightarrow \mu^{6-d} g$. In QED $[\mu] = 1$ and the coupling e is dimensionless for all d . Keeping g dimensionless at all d will produce logs with dimensionful arguments (I found this out in an earlier draft). So I allow $[g]$ to vary. I find $[g]$ in d dimensions.

$$[\phi] = \frac{d}{2} - 1, \quad [\mathcal{L}] = d \quad (55)$$

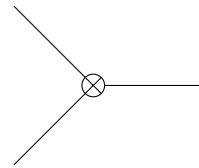
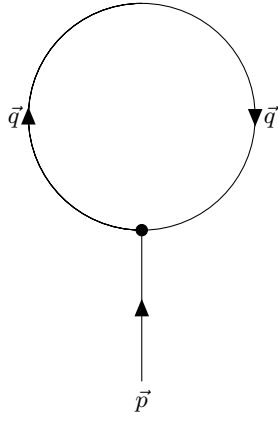


FIG. 3. Counterterm interaction vertex with factor $-i\delta_1 g$.


 FIG. 4. Amputated Feynman diagram $\mathfrak{F}$.

$$d = [\mathcal{L}] = [\mu^{6-d} g \phi^3] = [\mu^{6-d}] + [g] + [\phi^3] = (6-d)[\mu] + [g] + 3[\phi] = (6-d)[\mu] + [g] + 3\left(\frac{d}{2} - 1\right) \Rightarrow$$

$$[g] = d - 3\left(\frac{d}{2} - 1\right) - (6-d)[\mu] \quad (56)$$

I set $[\mu] = 1$ to keep log arguments dimensionless (Sec. 8.2).

$$[g] = \frac{d}{2} - 3 \quad (57)$$

7.1. Correction

Setting $[\mu] = 1$ keeps the 1-point log dimensionless but not the 2-point (Sec. 14.5). My convention in this homework is to use the rule $g \rightarrow \mu^{6-d} g$ and set $[\mu]$ for each Green's function to keep logs dimensionless. When calculating observables μ drops out and all logs are dimensionless.

8. 1-POINT GREEN'S FUNCTION

$$\langle \Omega | \phi(\vec{x}) | \Omega \rangle = \frac{i}{(2\pi)^6} \int d^6 p e^{\pm i \vec{p} \cdot \vec{x}} \mathcal{G}(\vec{p}) \quad (58)$$

The choice of $\pm$ is conventional. To lowest order there is one diagram (Fig. 4).

$$i \mathcal{G}(\vec{p}) = \frac{i \mathfrak{F}}{p^2 - m^2} \Rightarrow \mathcal{G}(\vec{p}) = \frac{\mathfrak{F}}{p^2 - m^2} \quad (59)$$

$$\mathfrak{F} = \frac{S}{(2\pi)^6} (-ig) \int d^6 q \frac{i}{q^2 - m^2 + i\epsilon} \quad (60)$$

S is the symmetry factor (Fig. 5).

$$S = \frac{3}{3!} = \frac{1}{2} \quad (61)$$

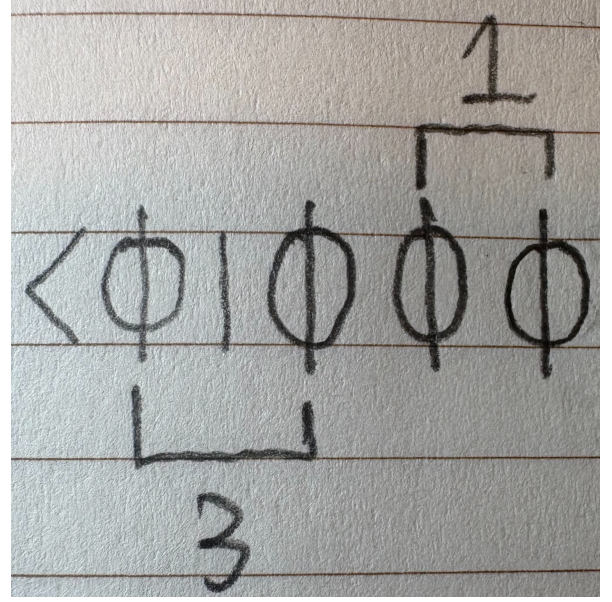


FIG. 5. Symmetry factor for Fig. 4.

$$\mathfrak{F} = \frac{g}{2(2\pi)^6} \int \frac{d^6 q}{q^2 - m^2 + i\epsilon} \quad (62)$$

8.1. UV Power Counting

$$\mathfrak{F} \sim \int_0^\Lambda dq \frac{q^5}{q^2} = \int_0^\Lambda dq q^3 \sim \Lambda^4, \quad \Lambda \rightarrow \infty \quad (63)$$

8.2. DimReg

$$\mathfrak{F}(d) = \frac{\mu^{6-d} g}{2(2\pi)^d} \int \frac{d^d q}{q^2 - m^2 + i\epsilon} \quad (64)$$

From my Real Scalar Field Notes

$$\frac{1}{(2\pi)^d} \int \frac{d^d p}{p^2 - m^2 + i\epsilon} = \frac{\Gamma(1 - d/2)}{i(4\pi)^{d/2} m^{2-d}}. \quad (65)$$

$$\mathfrak{F}(d) = \frac{\mu^{6-d} g}{2i(4\pi)^{d/2} m^{2-d}} \Gamma\left(1 - \frac{d}{2}\right) \quad (66)$$

$$\mathfrak{F} = \mathfrak{F}(6) = \mathfrak{F}(6 - \epsilon), \quad \epsilon \rightarrow 0 \quad (67)$$

$$\mathfrak{F} = \frac{\mu^\epsilon g}{2i(4\pi)^{3-\epsilon/2} m^{\epsilon-4}} \Gamma\left(\frac{\epsilon}{2} - 2\right) \quad (68)$$

$$\mathfrak{F} = \frac{gm^4}{2(4\pi)^3 i} \left(2\sqrt{\pi} \frac{\mu}{m}\right)^\epsilon \Gamma\left(\frac{\epsilon}{2} - 2\right) \quad (69)$$

$$\Gamma(x) = (x-1)\Gamma(x-1) \quad (70)$$

$$\gamma = \frac{gm^4}{2(4\pi)^3 i} \left(2\sqrt{\pi} \frac{\mu}{m}\right)^\epsilon \frac{\Gamma(\epsilon/2)}{(\epsilon/2-2)(\epsilon/2-1)} \quad (71)$$

$$\gamma = \frac{gm^4}{32\pi^3 i} \left(2\sqrt{\pi} \frac{\mu}{m}\right)^\epsilon \frac{\Gamma(\epsilon/2)}{(\epsilon-4)(\epsilon-2)} \quad (72)$$

I take $\epsilon \rightarrow 0$.

$$\gamma = \frac{gm^4}{128\pi^3 i} \left[\frac{1}{\epsilon} + \frac{3}{4} + \frac{1}{2} \ln \left(\frac{4\pi\mu^2}{e\gamma m^2} \right) \right] \quad (73)$$

To keep the log argument dimensionless I need $[\mu] = 1$ (Sec. 7).

$$\mathcal{G}(\vec{p}) = \frac{\gamma}{p^2 - m^2} \quad (74)$$

Momentum conservation at the vertex in Fig. 4 implies $\vec{p} = \vec{0} \Rightarrow p^2 = 0$.

$$\mathcal{G} = -\frac{\gamma}{m^2} \quad (75)$$

$$\mathcal{G} = \frac{igm^2}{128\pi^3} \left[\frac{1}{\epsilon} + \frac{3}{4} + \frac{1}{2} \ln \left(\frac{4\pi\mu^2}{e\gamma m^2} \right) \right] \quad (76)$$

No counterterms appear in this Green's function, so I cannot use it to fix any counterterm. I need to remove it with a field transformation.

9. SHIFTING THE FIELD

I remove the 1-point Green's function with the transformation

$$\Phi = \phi - \langle \phi \rangle \Rightarrow \phi = \Phi + \langle \phi \rangle, \quad \langle \phi \rangle = \langle \Omega | \phi(\vec{x}) | \Omega \rangle. \quad (77)$$

We do not need to do this in ϕ^4 theory or QED because in those QFTs $\langle \phi \rangle = \langle \vec{A} \rangle = \langle \psi \rangle = 0$.

10. NEW LAGRANGIAN

I derive the 1-loop Lagrangian for Φ .

$$\mathcal{L} = -\frac{\phi_B \square \phi_B}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!}, \quad \phi_B = \Phi_B + \langle \phi_B \rangle \quad (78)$$

$$\mathcal{L} = -\frac{\phi_B (\square \Phi_B + \square \langle \phi_B \rangle)}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!} \quad (79)$$

$\langle \phi_B \rangle$ is constant: $\square \langle \phi_B \rangle = 0$

$$\mathcal{L} = -\frac{\phi_B \square \Phi_B}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!} \quad (80)$$

$$\mathcal{L} = -\frac{\phi_B \square \Phi_B}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!} \quad (81)$$

$$\mathcal{L} = -\frac{(\Phi_B + \langle \phi_B \rangle) \square \Phi_B}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!} \quad (82)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{\langle \phi_B \rangle \square \Phi_B}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!} \quad (83)$$

I integrate by parts.

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} + \frac{\Phi_B \square \langle \phi_B \rangle}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!} \quad (84)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2 \phi_B^2}{2} - \frac{g_B \phi_B^3}{3!} \quad (85)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2}{2} (\Phi_B + \langle \phi_B \rangle)^2 - \frac{g_B \phi_B^3}{3!} \quad (86)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2}{2} \Phi_B^2 - \frac{m_B^2}{2} \langle \phi_B \rangle^2 - m_B^2 \Phi_B \langle \phi_B \rangle - \frac{g_B \phi_B^3}{3!} \quad (87)$$

Constant terms like $m_B^2 \langle \phi_B \rangle^2$ can be ignored.

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2}{2} \Phi_B^2 - m_B^2 \langle \phi_B \rangle \Phi_B - \frac{g_B \phi_B^3}{3!} \quad (88)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2}{2} \Phi_B^2 - m_B^2 \langle \phi_B \rangle \Phi_B - \frac{g_B}{3!} (\Phi_B + \langle \phi_B \rangle)^3 \quad (89)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2}{2} \Phi_B^2 - m_B^2 \langle \phi_B \rangle \Phi_B - \frac{g_B}{3!} (\Phi_B^3 + 3\langle \phi_B \rangle \Phi_B^2 + 3\langle \phi_B \rangle^2 \Phi_B + 3\langle \phi_B \rangle^3) \quad (90)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2}{2} \Phi_B^2 - m_B^2 \langle \phi_B \rangle \Phi_B - \frac{g_B}{3!} \Phi_B^3 - \frac{g_B}{2} \langle \phi_B \rangle^2 \Phi_B - \frac{g_B}{2} \langle \phi_B \rangle \Phi_B^2 \quad (91)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - (m_B^2 + g_B \langle \phi_B \rangle) \frac{\Phi_B^2}{2} - \left(m_B^2 + \frac{g_B}{2} \langle \phi_B \rangle \right) \langle \phi_B \rangle \Phi_B - \frac{g_B}{3!} \Phi_B^3 \quad (92)$$

The new field has a different mass.

$$M_B^2 = m_B^2 + g_B \langle \phi_B \rangle \quad (93)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{M_B^2}{2} \Phi_B^2 - \left(m_B^2 + \frac{g_B}{2} \langle \phi_B \rangle \right) \langle \phi_B \rangle \Phi_B - \frac{g_B}{3!} \Phi_B^3 \quad (94)$$

$$m_B^2 = M_B^2 - g_B \langle \phi_B \rangle \quad (95)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{M_B^2}{2} \Phi_B^2 - \left(M_B^2 - \frac{g_B}{2} \langle \phi_B \rangle \right) \langle \phi_B \rangle \Phi_B - \frac{g_B}{3!} \Phi_B^3 \quad (96)$$

$\langle \phi_B \rangle = c_B$ is a constant

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{M_B^2}{2} \Phi_B^2 - c_B \left(M_B^2 - \frac{g_B}{2} c_B \right) \Phi_B - \frac{g_B}{3!} \Phi_B^3 \quad (97)$$

$$C_B = -c_B \left(M_B^2 - \frac{g_B}{2} c_B \right) \quad (98)$$

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{M_B^2}{2} \Phi_B^2 + C_B \Phi_B - \frac{g_B}{3!} \Phi_B^3 \quad (99)$$

$$\Phi_B = \sqrt{Z_\Phi} \Phi, \quad M_B = \sqrt{Z_M} M, \quad g_B = Z_g g, \quad C_B = Z_C C \quad (100)$$

$$\mathcal{L} = -Z_\Phi \frac{\Phi \square \Phi}{2} - Z_M Z_\Phi \frac{M^2}{2} \Phi^2 + Z_C \sqrt{Z_\Phi} C \Phi - Z_g Z_\Phi^{3/2} \frac{g}{3!} \Phi^3 \quad (101)$$

$$Z_2 = Z_C \sqrt{Z_\Phi}, \quad Z_3 = Z_g Z_\Phi^{3/2} \quad (102)$$

$$\mathcal{L} = -Z_\Phi \frac{\Phi \square \Phi}{2} - Z_M Z_\Phi \frac{M^2}{2} \Phi^2 + Z_2 C \Phi - Z_3 \frac{g}{3!} \Phi^3 \quad (103)$$

$$Z_\Phi = 1 + \delta_\Phi, \quad Z_M = 1 + \delta_M, \quad Z_i = 1 + \delta_i, \quad i = 2, 3 \quad (104)$$

$$\mathcal{L} = -(1 + \delta_\Phi) \frac{\Phi \square \Phi}{2} - (1 + \delta_\Phi + \delta_M + \delta_M \delta_\Phi) \frac{M^2}{2} \Phi^2 + (1 + \delta_2) C \Phi - (1 + \delta_3) \frac{g}{3!} \Phi^3 \quad (105)$$

I go to 1-loop order.

$$\mathcal{L} = -(1 + \delta_\Phi) \frac{\Phi \square \Phi}{2} - (1 + \delta_\Phi + \delta_M) \frac{M^2}{2} \Phi^2 + (1 + \delta_2) C \Phi - (1 + \delta_3) \frac{g}{3!} \Phi^3 \quad (106)$$

$$\mathcal{L} = -\frac{\Phi(\square + M^2)\Phi}{2} + C\Phi - \frac{g}{3!} \Phi^3 - \frac{\delta_\Phi}{2} \Phi \square \Phi - (\delta_\Phi + \delta_M) \frac{M^2}{2} \Phi^2 + \delta_2 C \Phi - \delta_3 \frac{g}{3!} \Phi^3 \quad (107)$$

I redefine the free, interaction, and counterterm Lagrangians.

$$\boxed{2\mathcal{L}_0 = -\Phi(\square + M^2)\Phi}, \quad \boxed{\mathcal{L}_1 = C\Phi - \frac{g}{3!} \Phi^3} \quad (108)$$

$$\boxed{\mathcal{L}_C = -\frac{\Phi[\delta_\Phi \square + (\delta_\Phi + \delta_M)M^2]\Phi}{2} + \delta_2 C \Phi - \delta_3 \frac{g}{3!} \Phi^3} \quad (109)$$

$$\boxed{\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_C} \quad (110)$$



FIG. 6. One field interaction vertex with factor iC .

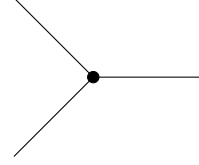


FIG. 7. Three field interaction vertex with factor $-ig$.

11. NEW FEYNMAN RULES

The propagator is

$$\Phi \sim \frac{i}{p^2 - M^2 + i\epsilon}. \quad (111)$$

The interaction vertices are in Figs. 6 & 7. The counterterm vertices in Figs. 8-10.

12. 1-POINT

$$\langle \Omega | \Phi(\vec{x}) | \Omega \rangle = \frac{i}{(2\pi)^6} \int d^6 p e^{\pm i\vec{p} \cdot \vec{x}} \mathcal{G}_1(\vec{p}) \quad (112)$$

To 1-loop order there are 3 diagrams (Figs. 11-13).

$$i\mathcal{G}_1(\vec{p}) = \mathfrak{F}[1] + \mathfrak{F}[2] + \mathfrak{F}[3] \quad (113)$$

Conservation of momentum implies $\vec{p} = \vec{0}$.

$$i\mathcal{G}_1 = \mathfrak{F}[1] + \mathfrak{F}[2] + \mathfrak{F}[3] \quad (114)$$

From Sec. 8.2

$$\mathfrak{F}[1] = \frac{i}{p^2 - M^2} \frac{gM^4}{128\pi^3 i} \left[\frac{1}{\epsilon} + \frac{3}{4} + \frac{1}{2} \ln \left(\frac{4\pi\mu^2}{e^\gamma M^2} \right) \right]. \quad (115)$$

$$\mathfrak{F}[1] = -\frac{gM^2}{128\pi^3} \left[\frac{1}{\epsilon} + \frac{3}{4} + \frac{1}{2} \ln \left(\frac{4\pi\mu^2}{e^\gamma M^2} \right) \right] \quad (116)$$

$$\boxed{\mathfrak{F}[1] = F - \frac{gM^2}{128\pi^3 \epsilon}}, \quad (117)$$

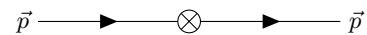
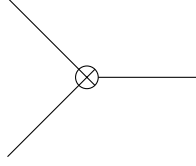


FIG. 8. Counterterm interaction vertex with factor $i[p^2 \delta_\Phi - (\delta_\Phi + \delta_M)M^2]$.


 FIG. 9. Factor $i\delta_2 C$

 FIG. 10. Counterterm interaction vertex with factor $-i\delta_3 g$.

F is the finite part

$$F = -\frac{gM^2}{128\pi^3} \left[\frac{3}{4} + \frac{1}{2} \ln \left(\frac{4\pi\mu^2}{e^\gamma M^2} \right) \right]. \quad (118)$$

$$\mathfrak{V}[2] = \frac{i}{p^2 - M^2} iC, \quad p^2 = 0 \quad (119)$$

$$\boxed{\mathfrak{V}[2] = \frac{C}{M^2}}, \quad \boxed{\mathfrak{V}[3] = \frac{C}{M^2} \delta_2} \quad (120)$$

$$i\mathcal{G}_1 = F - \frac{gM^2}{128\pi^3\epsilon} + \frac{C}{M^2} (1 + \delta_2) \quad (121)$$

$$\Phi_B = \phi_B - \langle \phi_B \rangle, \quad \langle \phi_B \rangle = \langle \Omega | \phi_B | \Omega \rangle \quad (122)$$

$$\sqrt{Z_\Phi} \Phi = \sqrt{Z_\phi} \phi - \sqrt{Z_\phi} \langle \phi \rangle \quad (123)$$

I assume $\sqrt{Z_\Phi} = \sqrt{Z_\phi}$, so

$$\Phi = \phi - \langle \phi \rangle. \quad (124)$$

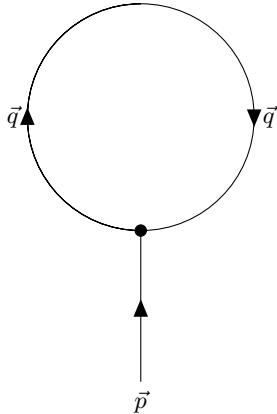

 FIG. 11. $\mathfrak{V}[1]$

 FIG. 12. $\mathfrak{V}[2]$

 FIG. 13. $\mathfrak{V}[3]$

$$\begin{aligned} \langle \Omega | \Phi | \Omega \rangle &= \langle \Omega | \phi - \langle \phi \rangle | \Omega \rangle = \langle \Omega | \phi | \Omega \rangle - \langle \Omega | \langle \phi \rangle | \Omega \rangle = \\ &= \langle \phi \rangle - \langle \phi \rangle \langle \Omega | \Omega \rangle = \langle \phi \rangle (1 - \langle \Omega | \Omega \rangle) \end{aligned} \quad (125)$$

$$\langle \Omega | \Omega \rangle = 1 \quad (126)$$

$$\langle \Omega | \Phi | \Omega \rangle = \langle \phi \rangle (1 - 1) = 0 \implies \mathcal{G}_1 = 0 \quad (127)$$

$$0 = F - \frac{gM^2}{128\pi^3\epsilon} + \frac{C}{M^2} (1 + \delta_2) \quad (128)$$

$$0 = F + \frac{C}{M^2} - \frac{gM^2}{128\pi^3\epsilon} + \frac{C}{M^2} \delta_2 \quad (129)$$

I use minimal subtraction to set the counterterms.

$$\frac{C}{M^2} \delta_2 - \frac{gM^2}{128\pi^3\epsilon} = 0 \implies \delta_2 = \frac{gM^4}{128\pi^3 C \epsilon} \quad (130)$$

With this subtraction scheme the renormalized linear coupling (to 1-loop order) is

$$F + \frac{C}{M^2} = 0 \implies \boxed{C = -FM^2}, \quad (131)$$

so

$$\boxed{\delta_2 = -\frac{gM^2}{128\pi^3 F \epsilon}}. \quad (132)$$

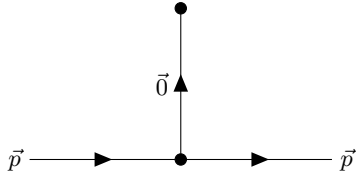
13. 2-POINT

$$\langle \Omega | T[\Phi(\vec{x}) \Phi(\vec{y})] | \Omega \rangle = \frac{i}{(2\pi)^6} \int d^6 p e^{i\vec{p} \cdot (\vec{y} - \vec{x})} \mathcal{G}_2(\vec{p}) \quad (133)$$

13.1. Tree Level

To lowest order there is one diagram (Fig. 14).

$$i\mathcal{G}^{(0)}(\vec{p}) = \frac{i}{p^2 - M^2} \implies \mathcal{G}^{(0)}(\vec{p}) = \frac{1}{p^2 - M^2} \quad (134)$$

FIG. 14. Feynman diagram for $i\mathcal{G}^{(0)}(\vec{p})$.FIG. 15. $\varphi[1]$

13.2. 1-Loop Order

The 1-loop correction has 5 diagrams.

$$i\Delta\mathcal{G}_2(\vec{p}) = \varphi[1] + \varphi[2] + \varphi[3] + \star + \mathfrak{C} \quad (135)$$

The tadpole (φ) diagrams are in Figs. 15 - 17. The counterterm ($\star$) diagram is in Fig. 18. The self-energy ($\mathfrak{C}$) diagram in Fig. 19.

$$\varphi = \varphi[1] + \varphi[2] + \varphi[3] \quad (136)$$

$$i\Delta\mathcal{G}_2(\vec{p}) = \varphi + \star + \mathfrak{C} \quad (137)$$

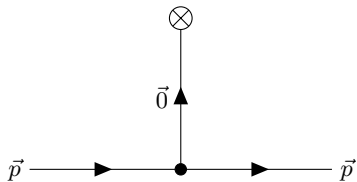
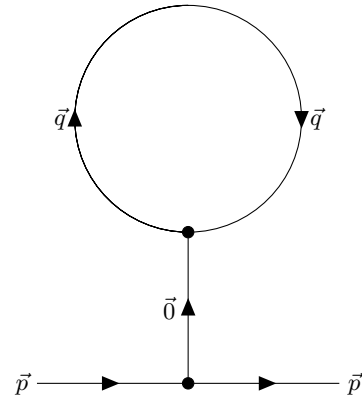
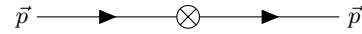
13.2.1. Tadpoles

$$\varphi[1] = \frac{i}{p^2 - M^2} (-ig) iC \frac{i}{0^2 - M^2} \frac{i}{p^2 - M^2} \quad (138)$$

$$\varphi[1] = \frac{i^5 g C}{(p^2 - M^2)^2 M^2} \quad (139)$$

$$\boxed{\varphi[1] = \frac{igC}{(p^2 - M^2)^2 M^2}} \quad (140)$$

$$\varphi[2] = \frac{i}{p^2 - M^2} (-ig) \frac{i}{0^2 - M^2} iC \delta_2 \frac{i}{p^2 - M^2} \quad (141)$$

FIG. 16. $\varphi[2]$ FIG. 17. $\varphi[3]$ FIG. 18. $\star$

$$\boxed{\varphi[2] = \frac{igC\delta_2}{(p^2 - M^2)^2 M^2}} \quad (142)$$

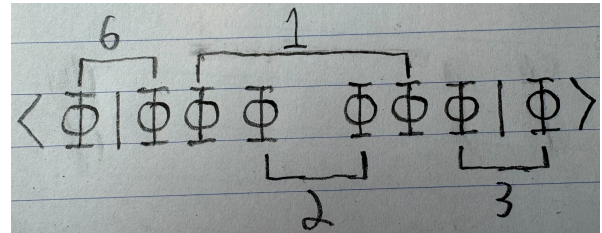
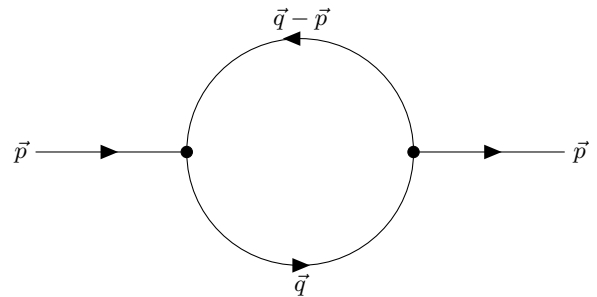
Equation 117 yields

$$\varphi[3] = \frac{i}{p^2 - M^2} (-ig) \left(F - \frac{gM^2}{128\pi^3\epsilon} \right) \frac{i}{p^2 - M^2}. \quad (143)$$

$$\boxed{\varphi[3] = \frac{ig}{(p^2 - M^2)^2} \left(F - \frac{gM^2}{128\pi^3\epsilon} \right)} \quad (144)$$

$$\varphi = \varphi[1] + \varphi[2] + \varphi[3] \quad (145)$$

$$\frac{(p^2 - M^2)^2}{ig} \varphi = \frac{C}{M^2} + \frac{C}{M^2} \delta_2 + F - \frac{gM^2}{128\pi^3\epsilon} \quad (146)$$

FIG. 19. Diagram $\mathfrak{C}$ with symmetry factor combinatorics.

$$\varphi \propto \frac{C}{M^2} + \frac{C}{M^2}\delta_2 + F - \frac{gM^2}{128\pi^3\epsilon} \quad (147)$$

$$\varphi \propto \frac{C}{M^2}(1 + \delta_2) + F - \frac{gM^2}{128\pi^3\epsilon} \quad (148)$$

Equations 131 & 132

$$C = -FM^2, \quad \delta_2 = -\frac{gM^2}{128\pi^3F\epsilon}, \quad (149)$$

yield

$$\varphi \propto -F\delta_2 - \frac{gM^2}{128\pi^3\epsilon} = \frac{gM^2}{128\pi^3\epsilon} - \frac{gM^2}{128\pi^3\epsilon} = 0 \implies \varphi = 0 \quad (150)$$

$$i\Delta\mathcal{G}_2(\vec{p}) = \star + \mathfrak{C} \quad (151)$$

The tadpole diagrams cancel. This is no coincidence; it's a consequence of my field shift in Sec. 9. It will happen for all Green's functions at all orders. So from now on I ignore tadpoles.

13.2.2. Counterterm Diagram

$$\star = \frac{i}{p^2 - M^2} i[p^2\delta_\Phi - (\delta_\Phi + \delta_M)M^2] \frac{i}{p^2 - M^2} \quad (152)$$

$$\star = i \frac{(\delta_\Phi + \delta_M)M^2 - p^2\delta_\Phi}{(p^2 - M^2)^2} \quad (153)$$

$$i\Delta\mathcal{G}_2(\vec{p}) = i \frac{(\delta_\Phi + \delta_M)M^2 - p^2\delta_\Phi}{(p^2 - M^2)^2} + \mathfrak{C} \quad (154)$$

13.2.3. Self-Energy Diagram

$$\mathfrak{C} = \frac{i}{p^2 - M^2} \Sigma \frac{i}{p^2 - M^2} \quad (155)$$

Σ is the amputated self-energy diagram.

$$\mathfrak{C} = -\frac{\Sigma}{(p^2 - M^2)^2} \quad (156)$$

14. AMPUTATED SELF-ENERGY

I evaluate Σ , the amputated diagram in Fig. 19.

$$\Sigma = \frac{S}{(2\pi)^6} (-ig)^2 \int d^6q \frac{i}{q^2 - M^2 + i\epsilon} \frac{i}{(\vec{q} - \vec{p})^2 - M^2 + i\epsilon} \quad (157)$$

The symmetry factor (S) is

$$S = \frac{6 \times 3 \times 2}{(3!)^2 2!} = \frac{1}{2}. \quad (158)$$

$$\Sigma = \frac{g^2}{2(2\pi)^6} \int \frac{d^6q}{(q^2 - M^2 + i\epsilon)[(\vec{q} - \vec{p})^2 - M^2 + i\epsilon]} \quad (159)$$

14.1. Feynman Parameters

$$A = (\vec{q} - \vec{p})^2 - M^2 + i\epsilon, \quad B = q^2 - M^2 + i\epsilon \quad (160)$$

$$\Sigma = \frac{g^2}{2(2\pi)^6} \int \frac{d^6q}{AB} \quad (161)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (162)$$

$$\Sigma = \frac{g^2}{2(2\pi)^6} \int d^6q \int_0^1 \frac{dx}{D^2} \quad (163)$$

$$D = (\vec{q} - \vec{p})^2 - M^2 + [q^2 - (\vec{q} - \vec{p})^2]x + i\epsilon \quad (164)$$

$$D = (\vec{q} - \vec{p})^2 + (2\vec{q} \cdot \vec{p} - p^2)x - M^2 + i\epsilon \quad (165)$$

$$D = q^2 - 2(1-x)\vec{q} \cdot \vec{p} + (1-x)p^2 - M^2 + i\epsilon \quad (166)$$

$$D = q^2 + (1-x)(p^2 - 2\vec{q} \cdot \vec{p}) - M^2 + i\epsilon \quad (167)$$

$$D = q^2 - 2(1-x)\vec{q} \cdot \vec{p} + (1-x)p^2 - M^2 + i\epsilon \quad (168)$$

14.2. Completing the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (169)$$

$$x \rightarrow \vec{q}, \quad a \rightarrow 1, \quad b \rightarrow -2(1-x)\vec{p}, \quad c \rightarrow (1-x)p^2 - M^2 + i\epsilon \quad (170)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - M^2 + x(1-x)p^2 + i\epsilon \quad (171)$$

$$\Delta = M^2 - x(1-x)p^2 \quad (172)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon \quad (173)$$

$$\Sigma = \frac{g^2}{2(2\pi)^6} \int d^6q \int_0^1 \frac{dx}{[\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon}^2 \quad (174)$$

14.3. Shift Integrand

$$\Sigma = \frac{g^2}{2(2\pi)^6} \int d^6 k \int_0^1 \frac{dx}{[\vec{k} - (1-x)\vec{p}]^2 - \Delta + i\epsilon} \quad (175)$$

$$\vec{q} = \vec{k} - (1-x)\vec{p}, \quad d^6 k = d^6 q \quad (176)$$

$$\Sigma = \frac{g^2}{2(2\pi)^6} \int d^6 q \int_0^1 \frac{dx}{(q^2 - \Delta + i\epsilon)^2} \quad (177)$$

$$\Sigma = \frac{g^2}{2(2\pi)^6} \int_0^1 dx \int \frac{d^6 q}{(q^2 - \Delta + i\epsilon)^2} \quad (178)$$

14.4. Divergence

The scalar mass M protects against IR divergences. There is a UV divergence:

$$\Sigma \sim \int_0^\Lambda dq \frac{q^5}{q^4} = \int_0^\Lambda dq q \sim \Lambda^2, \quad \Lambda \rightarrow \infty. \quad (179)$$

14.5. DimReg

$$\Sigma(d) = \frac{\mu^{2(6-d)} g^2}{2(2\pi)^d} \int_0^1 dx \int \frac{d^d q}{(q^2 - \Delta + i\epsilon)^2} \quad (180)$$

In my Real Scalar Field Notes I derived

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (181)$$

$$\Sigma(d) = \frac{i\mu^{12-2d} g^2}{2(2\pi)^d} \pi^{d/2} \Gamma\left(\frac{4-d}{2}\right) \int_0^1 \frac{dx}{\Delta^{2-d/2}} \quad (182)$$

$$\Sigma(d) = \frac{i\mu^{12-2d} g^2}{2(2\sqrt{\pi})^d} \Gamma\left(\frac{4-d}{2}\right) \int_0^1 \frac{dx}{\Delta^{2-d/2}} \quad (183)$$

$$\Sigma(d) = \frac{i\mu^{12} g^2}{2} \int_0^1 \frac{dx}{\Delta^2} \left(\frac{\sqrt{\Delta/\pi}}{2\mu^2}\right)^d \Gamma\left(2 - \frac{d}{2}\right) \quad (184)$$

$$\Sigma = \Sigma(6) = \Sigma(6 - \epsilon), \quad \epsilon \rightarrow 0 \quad (185)$$

$$\Sigma = \frac{i}{2} \mu^{12} g^2 \int_0^1 \frac{dx}{\Delta^2} \left(\frac{\sqrt{\Delta/\pi}}{2\mu^2}\right)^6 \left(\frac{2\mu^2}{\sqrt{\Delta/\pi}}\right)^\epsilon \Gamma\left(\frac{\epsilon}{2} - 1\right) \quad (186)$$

$$\Sigma = \frac{ig^2}{128\pi^3} \int_0^1 dx \Delta \left(\frac{2\mu^2}{\sqrt{\Delta/\pi}}\right)^\epsilon \Gamma\left(\frac{\epsilon}{2} - 1\right) \quad (187)$$

$$\Sigma = \frac{ig^2}{64\pi^3} \int_0^1 dx \Delta \left(\frac{2\mu^2}{\sqrt{\Delta/\pi}}\right)^\epsilon \frac{\Gamma(\epsilon/2)}{\epsilon - 2} \quad (188)$$

I take $\epsilon \rightarrow 0$.

$$\Sigma = \frac{g^2}{64i\pi^3\epsilon} \int_0^1 dx \Delta - i\mathcal{F} \quad (189)$$

The finite part is

$$\mathcal{F} = \frac{g^2}{128\pi^3} \int_0^1 dx \Delta \left[1 + \ln\left(\frac{4\pi\mu^4}{e^\gamma \Delta}\right)\right]. \quad (190)$$

Because $[\Delta] = 2$, I must take $[\mu] = 1/2$ to keep the log argument dimensionless.

$$\begin{aligned} \int_0^1 dx \Delta &= \int_0^1 dx [M^2 - x(1-x)p^2] = \\ M^2 \int_0^1 dx - p^2 \int_0^1 dx x(1-x) &= M^2 - \frac{p^2}{6} \end{aligned} \quad (191)$$

$$\Sigma = ig^2 \frac{p^2 - 6M^2}{384\pi^3\epsilon} - i\mathcal{F} \quad (192)$$

15. FINISH 2-POINT CALCULATION; MASS & FIELD COUNTERTERMS

$$i\Delta\mathcal{G}_2(\vec{p}) = i \frac{(\delta_\Phi + \delta_M)M^2 - p^2\delta_\Phi}{(p^2 - M^2)^2} + \mathcal{C} \quad (193)$$

$$\mathcal{C} = -\frac{\Sigma}{(p^2 - M^2)^2}, \quad \Sigma = ig^2 \frac{p^2 - 6M^2}{384\pi^3\epsilon} - i\mathcal{F} \quad (194)$$

$$(p^2 - M^2)^2 i\Delta\mathcal{G}_2 = i[(\delta_\Phi + \delta_M)M^2 - p^2\delta_\Phi] - \Sigma \quad (195)$$

$$(p^2 - M^2)^2 \Delta\mathcal{G}_2 = (\delta_\Phi + \delta_M)M^2 - p^2\delta_\Phi - g^2 \frac{p^2 - 6M^2}{384\pi^3\epsilon} + \mathcal{F} \quad (196)$$

$$\Delta\mathcal{G}_2 \propto \left(\delta_\Phi + \delta_M + \frac{g^2}{64\pi^3\epsilon}\right) M^2 - \left(\delta_\Phi + \frac{g^2}{384\pi^3\epsilon}\right) p^2 + \mathcal{F} \quad (197)$$

Minimal subtraction yields

$$\delta_\Phi + \frac{g^2}{384\pi^3\epsilon} = 0 \implies \boxed{\delta_\Phi = -\frac{g^2}{384\pi^3\epsilon}}. \quad (198)$$

$$\delta_\Phi + \delta_M + \frac{g^2}{64\pi^3\epsilon} = 0 \implies \boxed{\delta_M = -\frac{5g^2}{384\pi^3\epsilon}} \quad (199)$$

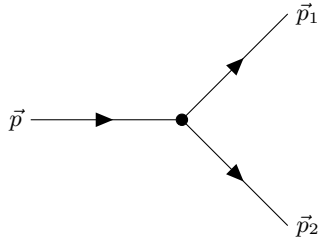


FIG. 20. ✱

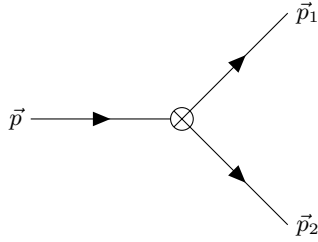


FIG. 21. ♣

16. 3-POINT

Instead of calculating the full 3-point Green's function I compute the amputated function (G) and get the counterterm with less algebra.

16.1. Tree Level

There is one tree level diagram (✱ in Fig. 20). Momentum conservation implies

$$\vec{p} = \vec{p}_1 + \vec{p}_2. \quad (200)$$

The lowest order is

$$iG^{(0)} = \star. \quad (201)$$

$$\boxed{\star = -ig}, \quad \boxed{G^{(0)} = -g} \quad (202)$$

16.2. Kinematics

In the frame where $\vec{p} = (M, 0, 0, 0, 0)$, a $1 \rightarrow 2$ decay obviously violates momentum conservation: that's okay for a Green's function, but the external momenta are off-shell.

$$p^2 = (\vec{p}_1 + \vec{p}_2)^2 \implies 2\vec{p}_1 \cdot \vec{p}_2 = p^2 - p_1^2 - p_2^2 \quad (203)$$

16.3. 1-Loop Order Correction

The leading correction has 2 diagrams (Figs. 21 & 22)

$$i\Delta G = \clubsuit + \spadesuit. \quad (204)$$

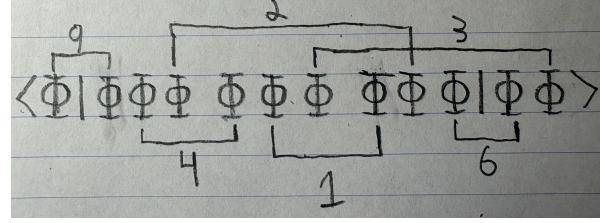
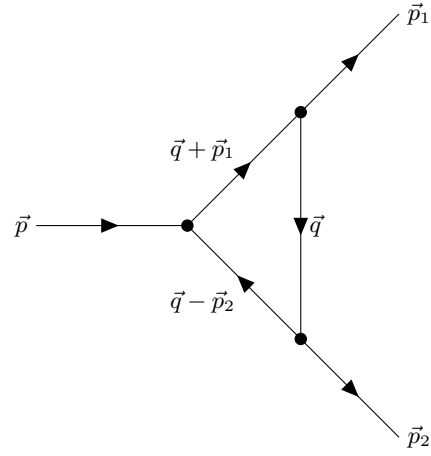


FIG. 22. Diagram ♠ with symmetry combinatorics.

16.3.1. Counterterm Diagram

$$\boxed{\clubsuit = -i\delta_3 g} \quad (205)$$

17. TRIANGLE LOOP

$$\spadesuit = \frac{S}{(2\pi)^6} (-ig)^3 \int d^6 q \frac{i}{q^2 - M^2 + i\epsilon} \frac{i}{(\vec{q} + \vec{p}_1)^2 - M^2 + i\epsilon} \frac{i}{(\vec{q} - \vec{p}_2)^2 - M^2 + i\epsilon} \quad (206)$$

The symmetry factor (S) combinatorics are in Fig. 22

$$S = \frac{9 \times 6 \times 4 \times 3 \times 2}{(3!)^3 3!} = 1. \quad (207)$$

$$\spadesuit = \frac{g^3}{(2\pi)^6} \int \frac{d^6 q}{q^2 - M^2 + i\epsilon} \frac{1}{[(\vec{q} + \vec{p}_1)^2 - M^2 + i\epsilon][(\vec{q} - \vec{p}_2)^2 - M^2 + i\epsilon]} \quad (208)$$

17.1. Feynman Parameters

$$A = q^2 - M^2 + i\epsilon, \quad B = (\vec{q} + \vec{p}_1)^2 - M^2 + i\epsilon, \quad C = (\vec{q} - \vec{p}_2)^2 - M^2 + i\epsilon \quad (209)$$

$$\spadesuit = \frac{g^3}{(2\pi)^6} \int \frac{d^6 q}{ABC} \quad (210)$$

$$\frac{1}{ABC} = 2 \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{D^3}, \quad D = xA + yB + zC \quad (211)$$

$$\spadesuit = \frac{2g^3}{(2\pi)^6} \int d^6 q \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{D^3} \quad (212)$$

$$D = xq^2 + y(\vec{q} + \vec{p}_1)^2 + z(\vec{q} - \vec{p}_2)^2 - (x+y+z)M^2 + (x+y+z)i\epsilon \quad (213)$$

The delta function implies $x + y + z = 1$.

$$D = xq^2 + y(\vec{q} + \vec{p}_1)^2 + z(\vec{q} - \vec{p}_2)^2 - M^2 + i\epsilon \quad (214)$$

$$D = xq^2 + y(q^2 + p_1^2 + 2\vec{q} \cdot \vec{p}_1) + z(q^2 + p_2^2 - 2\vec{q} \cdot \vec{p}_2) - M^2 + i\epsilon \quad (215)$$

$$D = (x+y+z)q^2 + 2(y\vec{p}_1 - z\vec{p}_2) \cdot \vec{q} + yp_1^2 + zp_2^2 - M^2 + i\epsilon \quad (216)$$

$$D = q^2 + 2(y\vec{p}_1 - z\vec{p}_2) \cdot \vec{q} + yp_1^2 + zp_2^2 - M^2 + i\epsilon \quad (217)$$

17.2. Completing the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (218)$$

$$x \rightarrow \vec{q}, \quad a \rightarrow 1, \quad b \rightarrow 2(y\vec{p}_1 - z\vec{p}_2), \quad c \rightarrow yp_1^2 + zp_2^2 - M^2 + i\epsilon \quad (219)$$

$$D = (\vec{q} + y\vec{p}_1 - z\vec{p}_2)^2 + yp_1^2 + zp_2^2 - (y\vec{p}_1 - z\vec{p}_2)^2 - M^2 + i\epsilon \quad (220)$$

$$D = (\vec{q} + y\vec{p}_1 - z\vec{p}_2)^2 + y(1-y)p_1^2 + z(1-z)p_2^2 + 2yz\vec{p}_1 \cdot \vec{p}_2 - M^2 + i\epsilon \quad (221)$$

$$2\vec{p}_1 \cdot \vec{p}_2 = p^2 - p_1^2 - p_2^2 \quad (222)$$

$$D = (\vec{q} + y\vec{p}_1 - z\vec{p}_2)^2 + (1-y-z)(yp_1^2 + zp_2^2) + yzp^2 - M^2 + i\epsilon \quad (223)$$

$$1 - y - z = x \quad (224)$$

$$D = (\vec{q} + y\vec{p}_1 - z\vec{p}_2)^2 - \heartsuit + i\epsilon \quad (225)$$

$$\heartsuit = M^2 - xyp_1^2 - xzp_2^2 - yzp^2 \quad (226)$$

$$\spadesuit = \frac{2g^3}{(2\pi)^6} \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^6 q}{[(\vec{q} + y\vec{p}_1 - z\vec{p}_2)^2 - \heartsuit + i\epsilon]^3} \quad (227)$$

17.3. Shift Integrand

$$\spadesuit = \frac{2g^3}{(2\pi)^6} \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^6 k}{[(\vec{k} + y\vec{p}_1 - z\vec{p}_2)^2 - \heartsuit + i\epsilon]^3} \quad (228)$$

$$\vec{q} = \vec{k} + y\vec{p}_1 - z\vec{p}_2, \quad d^6 k = d^6 q \quad (229)$$

$$\spadesuit = \frac{2g^3}{(2\pi)^6} \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^6 q}{(q^2 - \heartsuit + i\epsilon)^3} \quad (230)$$

$$I = \frac{g^3}{(2\pi)^6} \int \frac{d^6 q}{(q^2 - \heartsuit + i\epsilon)^3} \quad (231)$$

$$\spadesuit = 2 \int_0^1 dx dy dz \delta(x+y+z-1) I \quad (232)$$

17.4. Divergence

Mass prevents IR divergence. There is a UV log divergence

$$I \sim \int_1^\Lambda dq \frac{q^5}{q^6} = \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty. \quad (233)$$

17.5. DimReg

$$I(d) = \frac{\mu^{3(6-d)} g^3}{(2\pi)^d} \int \frac{d^d q}{(q^2 - \heartsuit + i\epsilon)^3} \quad (234)$$

From my Real Scalar Field Notes

$$\frac{1}{(2\pi)^d} \int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^b} = \frac{i(-1)^b}{(4\pi)^{d/2} \Delta^{b-d/2}} \frac{\Gamma(b-d/2)}{\Gamma(b)}. \quad (235)$$

$$I(d) = \frac{i(-1)^3 \mu^{3(6-d)} g^3}{(4\pi)^{d/2} \heartsuit^{3-d/2}} \frac{\Gamma(3-d/2)}{\Gamma(3)} \quad (236)$$

$$I(d) = \frac{\mu^{3(6-d)} g^3}{2i(4\pi)^{d/2} \heartsuit^{3-d/2}} \Gamma\left(3 - \frac{d}{2}\right) \quad (237)$$

$$I(d) = \frac{\mu^{18} g^3}{2i\heartsuit^3} \left(\frac{\sqrt{\heartsuit/\pi}}{2\mu^3} \right)^d \Gamma\left(3 - \frac{d}{2}\right) \quad (238)$$

$$I = I(6) = I(6 - \epsilon), \quad \epsilon \rightarrow 0 \quad (239) \quad \text{which yields}$$

$$I = \frac{\mu^{18} g^3}{2i\heartsuit^3} \left(\frac{\sqrt{\heartsuit/\pi}}{2\mu^3} \right)^{6-\epsilon} \Gamma\left(\frac{\epsilon}{2}\right) \quad (240)$$

$$I = \frac{g^3}{128i\pi^3} \left(\frac{2\mu^3}{\sqrt{\heartsuit/\pi}} \right)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \quad (241)$$

I take $\epsilon \rightarrow 0$.

$$I = \frac{g^3}{128i\pi^3} \left[\frac{2}{\epsilon} - \gamma + \ln\left(4\pi \frac{\mu^6}{\heartsuit}\right) \right] \quad (242)$$

Keeping the log dimensionless (Sec. 7.1) requires

$$6[\mu] = [\heartsuit] = 2 \implies [\mu] = \frac{1}{3}. \quad (243)$$

$$\mathfrak{U} = \int_0^1 dx \int_0^{1-x} dy = \int_0^1 dx (1-x) = \frac{1}{2}. \quad (252)$$

$$\spadesuit = \frac{g^3}{64i\pi^3} \left[\frac{1}{\epsilon} - \frac{\gamma}{2} + \int_0^1 dx dy dz \delta(x+y+z-1) \ln\left(4\pi \frac{\mu^6}{\heartsuit}\right) \right] \quad (253)$$

$$\spadesuit = i\mathbb{F} - \frac{ig^3}{64\pi^3\epsilon} \quad (254)$$

$$\frac{64\pi^3}{g^3} \mathbb{F} = \frac{\gamma}{2} - \int_0^1 dx dy dz \delta(x+y+z-1) \ln\left(4\pi \frac{\mu^6}{\heartsuit}\right) \quad (255)$$

17.6. Final

18. FINAL COUNTERTERM

$$\spadesuit = 2 \int_0^1 dx dy dz \delta(x+y+z-1) I \quad (244)$$

$$I = \frac{g^3}{128i\pi^3} \left[\frac{2}{\epsilon} - \gamma + \ln\left(4\pi \frac{\mu^6}{\heartsuit}\right) \right] \quad (245)$$

$$\spadesuit = \frac{g^3}{64i\pi^3} \left[\left(\frac{2}{\epsilon} - \gamma \right) \mathfrak{U} + \int_0^1 dx dy dz \delta(x+y+z-1) \ln\left(4\pi \frac{\mu^6}{\heartsuit}\right) \right] \quad (246)$$

$$\mathfrak{U} = \int_0^1 dx dy dz \delta(x+y+z-1) \quad (247)$$

$$\mathfrak{U} = \int_0^1 dx \int_0^1 dy \int_0^1 dz \delta(x+y+z-1) \quad (248)$$

$$\mathfrak{U} = \int_0^1 dx \int_0^1 dy \int_0^1 dz \delta[z - (1-x-y)] \quad (249)$$

Because $0 \leq z \leq 1$, $z = 1-x-y \implies 0 \leq 1-x-y \leq 1$.
Because $0 \leq x, y \leq 1$ the right hand condition is satisfied

$$1-x-y \leq 1 \implies -x-y \leq 0 \implies \text{TRUE}. \quad (250)$$

The left hand condition implies

$$0 \leq 1-x-y \implies x+y \leq 1 \implies y \leq 1-x, \quad (251)$$

$$i\Delta G = \clubsuit + \spadesuit, \quad \clubsuit = -i\delta_3 g, \quad \spadesuit = i\mathbb{F} - \frac{ig^3}{64\pi^3\epsilon} \quad (256)$$

$$i\Delta G = i\mathbb{F} - \frac{ig^3}{64\pi^3\epsilon} - i\delta_3 g \quad (257)$$

$$\Delta G = \mathbb{F} - \frac{g^3}{64\pi^3\epsilon} - \delta_3 g \quad (258)$$

$$\frac{g^3}{64\pi^3\epsilon} + \delta_3 g = 0 \implies \delta_3 = -\frac{g^2}{64\pi^3\epsilon} \quad (259)$$